Yanlin Chen

yanlinc2@andrew.cmu.edu 1-412-287-4214
Website: https://yl10016.github.io/yanlin.github.io/

Education

•Carnegie Mellon University QPA: 3.9/4.0
B.S. in Computer Science (Concentration in ML and Minor in Mathematics) *Aug 2023 – Aug 2026*

Work Experience

•Network Development Intern, Amazon, Cupertino *May 2026 - July 2026*
- Optics

•Network Product (SWE) Intern, Cisco, San Jose *May 2025 - Aug 2025*
- Developed AI-driven, **remotely hosted** and **Dockerized Flask application** to automate bug triage and root cause analysis (RCA) by retrieving and correlating cluster, device, and codebase data using **LangChain agents** enhanced with API/CLI tools, MCP servers, local regex search, and token usage tracking (and general traces monitoring) via **Langfuse**
- Integrated the solution as a **VSCode extension** to enable Copilot-powered, inline code fix suggestions directly within the developer workflow, and implemented fast similarity search of existing CDETS bugs using a Milvus vector database with TF-IDF and cosine matching to surface relevant existing issues

•Physics Instructor at NUS High School of Math and Science *2021 - 2023*
- Trained a select group of students in preparation for national physics research tournaments: coached students in physics, mathematical modelling, programming, electronic circuit and mechanical setup construction and design
- Taught participants Mathematica and Computational Fluid Dynamics software such as COMSOL and Ansys

Technical Skills: GoLang, Python, C, Standard ML, Linux/Unix, Qiskit, Mathematica

Research Experience

•Chamanzar Group, Carnegie Mellon University *Sept 2024 - Present*
- Designed a microwave antenna (via Ansys HFSS simulations) to drive spin transitions in SiC quantum photonic devices, to detect weak magnetic fields generated by neural activity using the Zeeman splitting effect
- Characterized Optically Detected Magnetic Resonance (ODMR) spectra by analyzing resonance dip positions (by sweeping microwave frequency, and recording fluorescence intensity changes between microwave-on/off conditions)

•Institute of High Performance Computing, A*STAR *May 2024 - July 2024*
- Built and ran circuits in Qiskit to develop and evaluate a zero-noise extrapolation (ZNE) approach using the Parameter Shift Rule (PSR), to perform error mitigation without increasing the circuit depth

Clubs and Projects

•Kiltie Band (Marching, Concert, and Pep Band) *Aug 2023 - Present*
- Member of CMU's marching, concert, and pep band (Clarinet section) for all 6 semesters

•Hackberry Pi (Hardware) Hackathon *March 2026*
- Engineered an interactive aircraft hologram system via a 3D frustum projection pipeline that maps OpenSky API data onto perspective-correct views, with joystick/controller input for live map panning, won 2nd place in the general track

•Distributed Systems (in Go) *Sept 2025 - Dec 2025*
- Developed a large-scale, multi-component system that communicated via **Remote Procedure Calls** and used the **Actor Model** to coordinate work across cores and machines, enforcing consistency guarantees for replicated data
- Implemented the **Raft consensus algorithm**, a replicated state machine protocol that serves as an alternative to Paxos, enabling leader election, log replication, and consistent fault-tolerant state across distributed nodes
- Built a custom reliable transport protocol (**Live Sequence Protocol**) on top of UDP that leveraged goroutines and sliding-window mechanisms to ensure ordered delivery, integrity checks, exponential backoff retransmissions, and fault-tolerant client–server coordination for parallel hashing tasks (used in a distributed bitcoin miner scenario)

•Jane Street INSIGHT Program *Jan 2025*
- Built a trading algorithm and traded in a virtual exchange (as one of 30 participants selected for the SWE track)

•Computer Systems and Programs (in C) *Aug 2024 - Dec 2024*
- Developed a proxy (compatible with Mozilla Firefox), shell (back/foreground job control and I/O redirection), malloc (best fit with segregated lists), and cache (LRU) simulator totaling around 4000 lines

Achievements

•Jane Street Market Making Challenge *2026*
Ranked third in the Estimation segment across 40 teams

•CMU MSCF (Master of Science in Computational Finance) Datathon *2025*
Trained an AI trading bot on institutional financial data, ranking #4 overall and #2 in Pittsburgh across 18 active teams

•Gold/Champion at (Online) International Young Physicist Tournament *2020, 2021*
Member of Singapore's national team at international research-focused competitions